# MA1522 Linear Algebra for Computing

## Lesson 4: Inverses, Elementary Matrices and Their Uses

# Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson4` (e.g. `A1234567Z_lesson4.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 4**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

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For final:

No MATLAB

Calculator only.

List of approved Calculators

– see Canvas

Basic rules:

- Please (mainly) ask the questions related to the course.

# Example 1

Solve the linear system

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ 2x_1 + 5x_2 + 3x_3 = 2 \\ x_1 + 8x_3 = 3 \end{cases}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 5 & 3 & 2 \\ 1 & 0 & 8 & 3 \end{array}\right) \xrightarrow{\text{ref}} \left(\begin{array}{ccc|c} 1 & & & 19 \\ & 1 & & -6 \\ & & 1 & -2 \end{array}\right)$$

$$\Rightarrow \quad x_1 = 19, \quad x_2 = -6, \quad x_3 = -2$$

# Section 2.4 Inverse of Matrices

A square matrix $A$ of order $n$ is **invertible** if there is a matrix $B$ with

$$AB = I_n = BA \qquad \text{(both products, and then } B \text{ is } n \times n \text{ too).}$$

Such a $B$ is unique — if $AB = I = BA$ and $AC = I = CA$ then $B = B(AC) = (BA)C = C$ — so we may write $B = A^{-1}$, **the** inverse of $A$. A square matrix with no inverse is called **singular**.

### Remark

A matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of order 2 is invertible if and only if $ad - bc \neq 0$, and in that case

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The generalisation of this formula to order $n$ is called **Cramer's formula**, and we will meet it in Section 2.10.

# Section 2.4 Properties of the Inverse, and How to Find It

Let $A$, $B$ be invertible of order $n$ and $c \neq 0$.

- $(A^{-1})^{-1} = A$;
- $(cA)^{-1} = \frac{1}{c}A^{-1}$;
- $(A^T)^{-1} = (A^{-1})^T$;
- $(AB)^{-1} = B^{-1}A^{-1}$ — the order reverses, exactly as for the transpose;
- $A^{-n} := (A^{-1})^n$.

$(B^{-1}A^{-1})\cdot AB = I_n$.

$A^{-1}A = I_n$, $B^{-1}I_n = I_n$

**The algorithm.** Finding $A^{-1}$ means solving $AX = I_n$, which by block multiplication is $n$ systems with the *same* coefficient matrix. So run Gauss–Jordan once, on $(A \mid I_n)$:

$$(A \mid I) \xrightarrow{\text{Gauss–Jordan}} (I \mid A^{-1})$$

$n \times 2n$; rref.

If a zero row ever appears in the left block, stop: the RREF of $A$ is not $I$, and $A$ is singular.

# Question 1

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ — the coefficient matrix and the constant vector of Example 1.

1. Find $A^{-1}$. by the algorithm $(A \mid I) \xrightarrow{\text{ref}} (I \mid A^{-1})$
2. Hence solve $A\mathbf{x} = \mathbf{b}$.
3. Without any further row reduction, write down $(A^T)^{-1}$ and $(3A)^{-1}$. from $A^{-1}$. $(A^{-1})^T$ $\frac{1}{3} \cdot A^{-1}$

# Question 1 — solution: part (1)

(1) $\left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right) \xrightarrow{\text{rref}} \left(\begin{array}{ccc|ccc} 1 & & & -40 & 16 & 9 \\ & 1 & & 13 & -5 & -3 \\ & & 1 & 5 & -2 & -1 \end{array}\right)$

$\Rightarrow A^{-1} = \begin{pmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{pmatrix}$

(2) Solve $Ax = b = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.

$ax = b \quad a, b \in \mathbb{R}$.

$a^{-1} a x = a^{-1} b$

$x = \frac{b}{a}$.

$A^{-1}(Ax) = A^{-1} b$

$I_3 \quad x$

$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x$

$= \begin{pmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 19 \\ -6 \\ 2 \end{pmatrix}$.

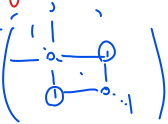
# Question 1 — solution: parts (2) and (3)

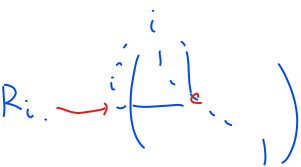
Rmk.

If we have $A^{-1}$ then we can solve also solving $Ax = \begin{pmatrix}1\\0\\\vdots\\0\end{pmatrix}$, $Ax = \begin{pmatrix}0\\1\\\vdots\\0\end{pmatrix}$, $Ax = \begin{pmatrix}0\\0\\\vdots\\1\end{pmatrix}$.

$$Ax = b \quad \text{by} \quad x = A^{-1}b.$$

elementary matrix.

$R_i \leftrightarrow R_j$. $\rightsquigarrow$ 

$cR_i$. $\rightsquigarrow$ 

$R_i + aR_j$.

# Section 2.5 Elementary Matrices

An **elementary matrix** $E$ is what you get by doing one elementary row operation to $I_n$. The point of them: $A \xrightarrow{r} EA$ — doing $r$ to $A$ = premultiplying $A$ by $E$.

Here is an example.

$$EA = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \mathbf{r}_3 \end{pmatrix} = \begin{pmatrix} 1\mathbf{r}_1 + 0\mathbf{r}_2 + 0\mathbf{r}_3 \\ 3\mathbf{r}_1 + 1\mathbf{r}_2 + 0\mathbf{r}_3 \\ 0\mathbf{r}_1 + 0\mathbf{r}_2 + 1\mathbf{r}_3 \end{pmatrix} = \begin{pmatrix} \mathbf{r}_1 \\ 3\mathbf{r}_1 + \mathbf{r}_2 \\ \mathbf{r}_3 \end{pmatrix} \quad \text{— exactly } R_2 + 3R_1.$$

- Every elementary matrix is invertible, and $E^{-1}$ is the elementary matrix of the **reverse** operation — here $E^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, the matrix of $R_2 - 3R_1$:

$$R_i + cR_j \;\leftrightarrow\; R_i - cR_j, \qquad R_i \leftrightarrow R_j \text{ is its own reverse}, \qquad cR_i \;\leftrightarrow\; \tfrac{1}{c}R_i.$$

- Hence if $A \xrightarrow{r_1} \cdots \xrightarrow{r_k} B$ with elementary matrices $E_1, \ldots, E_k$, then

$$B = E_k \cdots E_1 A \qquad \text{and} \qquad A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} B.$$

Both orders reverse. Read the second one as: undo $r_k$ first.

# Question 2

A $3 \times 3$ matrix $A$ and a vector $\mathbf{b}$ are not given to you. All you are told is what three row operations do to them:

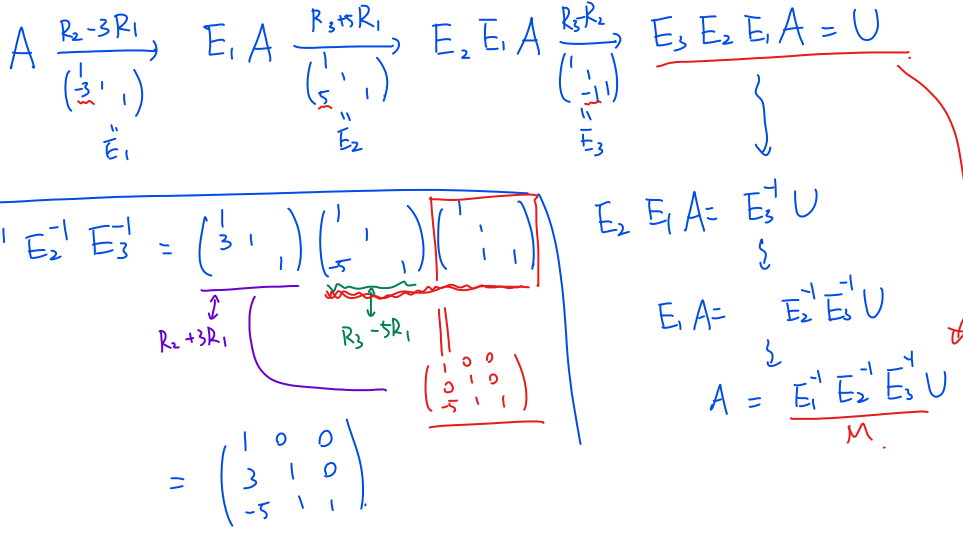
$$\begin{pmatrix} ? & ? & ? \\ ? & ? & ? \\ ? & ? & ? \end{pmatrix} = A \xrightarrow{R_2 - 3R_1} \xrightarrow{R_3 + 5R_1} \xrightarrow{R_3 - R_2} U = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{pmatrix},$$

$$\begin{pmatrix} ? \\ ? \\ ? \end{pmatrix} = \mathbf{b} \xrightarrow{R_2 - 3R_1} \xrightarrow{R_3 + 5R_1} \xrightarrow{R_3 - R_2} \mathbf{c} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

1. Find a matrix $M$ with $A = MU$.
2. Hence solve $A\mathbf{x} = \mathbf{b}$, still without working out what $A$ and $\mathbf{b}$ are.
3. Now work out $A$ and $\mathbf{b}$, and check your answer to (2).

Step 1. Write down the elmt matrix for these operations and their inverse.

# Question 2 — solution: part (1)



# Question 2 — solution: parts (2) and (3)

(2) Solve $Ax = b$ given $A = MU$

$b \longrightarrow \longrightarrow \longrightarrow c = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$.

$MUx = Mc \Rightarrow M^{-1}MUx = M^{-1}Mc$

(☆) $Ux = c$ $\qquad b = Mc$.

$U = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{pmatrix}$ $\quad c = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. $\quad x = \begin{pmatrix} -\frac{1+t}{2} - 2t \\ \frac{1+t}{2} \\ t \end{pmatrix}$ $\quad t \in \mathbb{R}$.

(3) $A = MU$ $\qquad b = Mc$

# Section 2.6 Equivalent Statements for Invertibility

## Theorem

For a square matrix $A$ of order $n$, the following are **equivalent** — each one implies all the others.

1. $A$ is invertible;
2. $A^T$ is invertible;
3. $BA = I$ for some $B$ (a **left inverse**);
4. $AB = I$ for some $B$ (a **right inverse**);
5. the RREF of $A$ is $I_n$;
6. $A$ is a product of elementary matrices;
7. $A\mathbf{x} = \mathbf{0}$ has only $\mathbf{x} = \mathbf{0}$;
8. $A\mathbf{x} = \mathbf{b}$ is uniquely solvable, for every $\mathbf{b}$.

## Challenge

Let $A$ and $B$ be square matrices of the same size. Show that if $AB = I$, then $BA = I$. (You may use the equivalence above.)

# Example 2

$(AB)^{-1} = B^{-1}A^{-1}$ if $A, B$ are invertible.

**True or false?** Let $A$ and $B$ be square matrices of the same size. If $AB$ is invertible, then $A$ and $B$ are both invertible.

True!

Let $C$ be the inverse of $AB$, i.e.

$$I = (AB)C = A(BC).$$

$BC$: right inverse of $A$ $\Rightarrow$ $A$ is invertible.

$$I = C(AB) = (CA)B.$$

$CA$: left inverse of $B$. $\Rightarrow$ $B$ is invertible.

# Question 3

Let $A$ be a $3 \times 3$ matrix, and suppose you are told only that each of the three systems

$$A\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad A\mathbf{x} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \qquad A\mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

is consistent. Which of the following must be true? Justify your answer.

1. $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution. $\Leftrightarrow$ $A$ is singular.
2. There is some $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions. $\Leftrightarrow$ $A$ is singular.
3. Every REF of $A$ has three nonzero rows. $\Leftrightarrow$ $\mathrm{rref}(A) = I$. $\Leftrightarrow$ $A$ is invertible.
4. There is not enough information to decide (1), (2) or (3).

# Question 3 — solution

Conditions :

$\exists\ u_1, u_2, u_3$ s.t. $Au_1 = \begin{pmatrix}1\\0\\1\end{pmatrix}$, $Au_2 = \begin{pmatrix}0\\1\\1\end{pmatrix}$, $Au_3 = \begin{pmatrix}1\\1\\0\end{pmatrix}$.

(Hint: use example 2).

$\Downarrow$ $U$

$$A \begin{pmatrix} | & | & | \\ u_1 & u_2 & u_3 \\ | & | & | \end{pmatrix} = \begin{pmatrix} | & | & | \\ Au_1 & Au_2 & Au_3 \\ | & | & | \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

invertible !

ref

Example 2 $\Rightarrow$ $A$ is invertible.

$\Rightarrow$ 3 is correct.

# Challenge

A square matrix $A$ is **nilpotent** if $A^k = 0$ for some $k \geq 1$; for instance $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ has $A^3 = 0$.

Show that for such an $A$, the matrix $I - A$ is invertible, and find $(I - A)^{-1}$.

# Next lesson

- **Sections 2.7–2.9 LU Factorization and Determinants**

Please watch the video clips for these sections beforehand.

If you do not know how to upload your submission, come to ask me after class.